

Wheatstone Bridge

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[Analog and Integrated Circuits: BM2013]

September 22, 2025

1 Introduction

1.1 Wheatstone Bridge: A Simple Tool for Measuring Resistance

A Wheatstone bridge is a classic electrical circuit consisting of four resistors arranged in a diamond configuration. By adjusting the resistors until the bridge is in balance, one can determine the value of an unknown resistance with high accuracy.

The idea is similar to a weighing balance: when both sides are in equilibrium, the comparison is precise. Likewise, in a Wheatstone bridge, balance occurs when the ratio of known resistances matches the ratio involving the unknown resistance.

Applications of Wheatstone Bridge include:

- **Strain measurement:** Detecting how a material deforms under stress.
- **Temperature sensing:** Determining changes in temperature through resistance variation.
- **General resistance measurement:** Used in laboratories and instrumentation to obtain accurate electrical values.

1.2 Objective:

Build a Wheatstone bridge circuit and see how it works. Compare the results to what we expect based on calculations. Then, make a visual representation of our findings.

2 Theory

2.1 Balanced Wheatstone Bridge Equation

For a balanced Wheatstone bridge, the ratio of resistances in one leg is equal to the ratio of resistances in the other leg:

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \quad (1)$$

$$R_2 \quad R_4$$

where:

R_1 and R_2 are the resistances in one leg

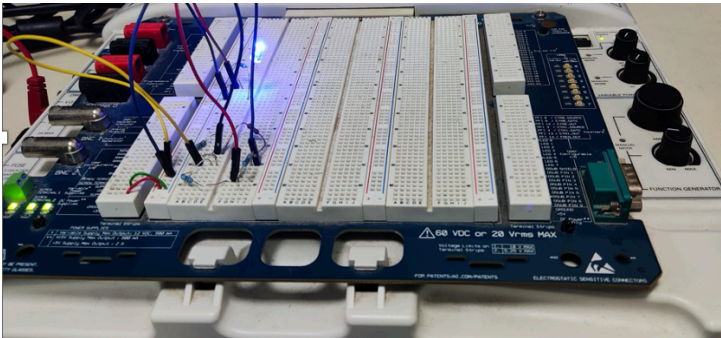
R_3 and R_4 are the resistances in the other leg

2.2 Significance of Wheatstone Bridge

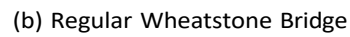
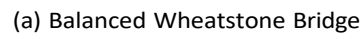
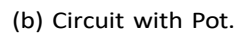
A balanced Wheatstone bridge is a valuable tool for precise electrical measurements. It's like a scale for electricity, helping us compare and measure resistance accurately.

3 Experimental Setup and Circuit Diagram

Components include 4 resistors, 1 Galvanometer, 1 Potentiometer, Elvis Board, PC, Required Software (TINA and ELVIS INS.), jumper wires.



(a) Wheatstone bridge on Elvis Board



4 Procedure

4.1 Balancing a Wheatstone Bridge

For a balanced setup, the variable resistor is tuned until the galvanometer shows no deflection, indicating zero current flow through the bridge branch.

In the unbalanced case, the resistances are intentionally mismatched so that the galvanometer shows a deflection, allowing us to study the potential difference across the bridge.

4.2 Finding Unknown Resistance (Balanced Wheatstone Bridge)

Once the bridge is balanced, the unknown resistance can be calculated using the formula:

$$\text{Unknown Resistance} = \frac{(\text{KnownResistance1} * \text{KnownResistance2})}{\text{KnownResistance3}}$$

This formula is based on the principle that in a balanced Wheatstone bridge, the ratio of resistances in one leg is equal to the ratio of resistances in the other leg.

4.2 Determining Unknown Resistance (Unbalanced Bridge with Potentiometer)

In the unbalanced Wheatstone bridge, a potentiometer R_p is placed in one branch. Let:

- R_1, R_3, R_4 : known resistors in the circuit
- V : applied supply voltage
- V_1, V_2 : potential drops across the two arms
- I : total current in the bridge

Applying Kirchhoff's Voltage Law (KVL):

$$V_1 = I \cdot (R_1 + R_p), V_2 = I \cdot (R_3 + R_4)$$

Since the bridge is not balanced, $V_1 \neq V_2$. The potential difference between the arms becomes:

$$\Delta V = V_1 - V_2 = I \cdot [(R_1 + R_p) - (R_3 + R_4)]$$

Rearranging for the potentiometer resistance:

$$R_p = \frac{\Delta V}{I} - R_1 + R_3 + R_4$$

This expression enables the calculation of the potentiometer resistance using experimental voltage readings, the total current, and the known resistor values.

5 Observations

5.1 Balanced Wheatstone Bridge

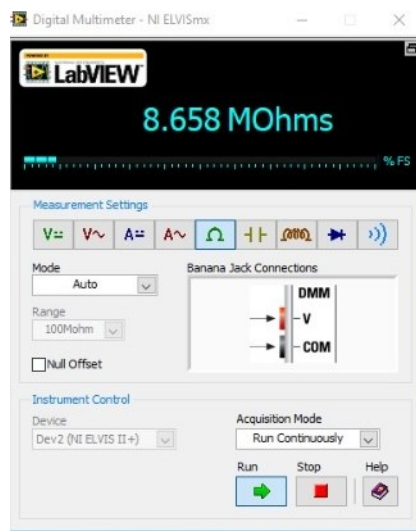
Readings of Resistance and Galvanometer:



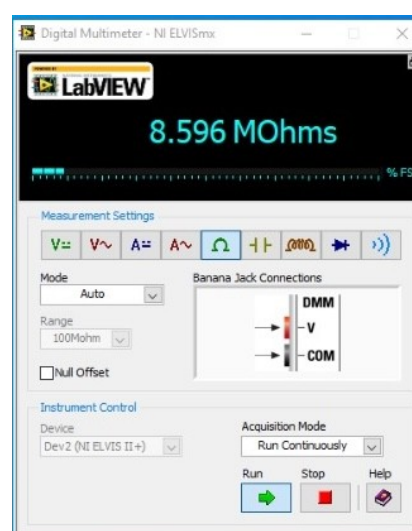
(a) Reading of Resistance 1



(b) Reading of Resistance 2



(a) Reading of Resistance 3



(b) Reading of Resistance 4

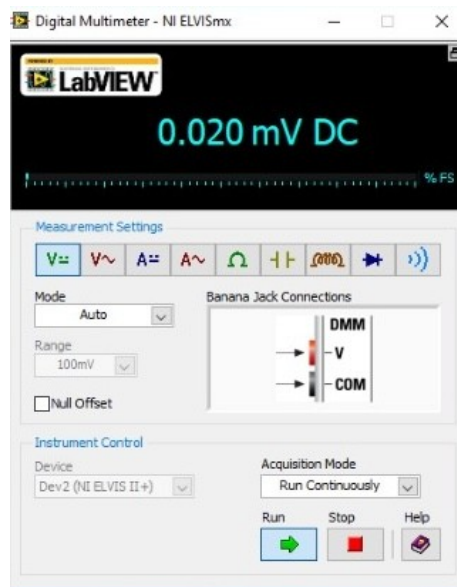


Figure 5: The voltmeter reading across the bridge was approximately 0.020 mV, suggesting a near-balanced Wheatstone bridge.

Calculated Value

S.No.	Resistor Value
Resistor 1	8.636
Resistor 2	8.541
Resistor 3	8.658
Resistor 4	8.596

5.2 Unbalanced Wheatstone Bridge

Different resistors were used for the unbalanced Wheatstone bridge (from balanced), with three resistors having the same value, and a potentiometer was used to vary the resistance of the fourth resistor. The resistors used had values of 0.32 Mohm, 0.30 Mohm, and 0.30 Mohm, respectively. Five readings were taken, indicating the resistance of the potentiometer and the voltage observed by the voltmeter. Negative resistance was observed

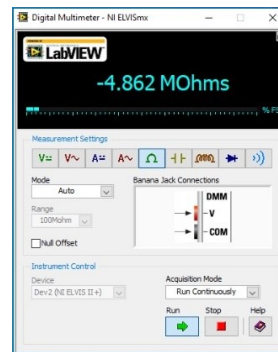
due to the opposite connection of the banana plugs/ Negative resistance was observed due to the reversed polarity.

Readings of Pot Resistance and Voltmeter:

Resistance observed/taken (Potentiometer)	Voltage observed (In Voltmeter)
4.862 Mohm	1.9309V
4.967 Mohm	2.1238V
4.995 Mohm	2.1562V
5.013 Mohm	2.1995V
5.089 Mohm	2.2362V



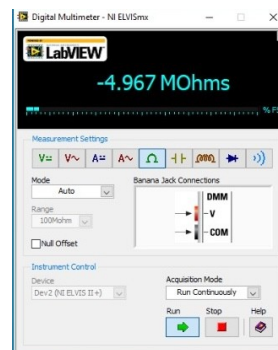
Voltmeter Reading 1



(b) Potentiometer Reading 1



(a) Voltmeter Reading 2



(b) Potentiometer Reading 2



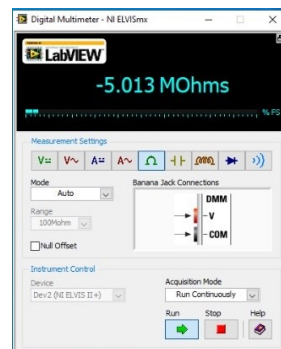
(a) Voltmeter Reading 3



(b) Potentiometer Reading 3



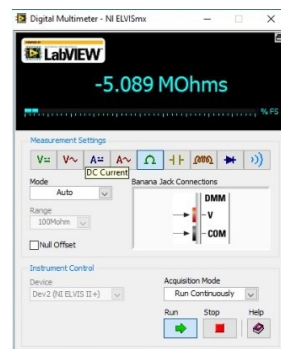
(a) Voltmeter Reading 4



(b) Potentiometer Reading 4



(a) Voltmeter Reading 5



(b) Potentiometer Reading 5

The observed reading was 2.2362 V, while the calculated reading was 2.1232 V. The close proximity of these values indicates that the experiment was conducted correctly. Similarly, we can verify the accuracy of all five resistance values.

5.3 Graph:

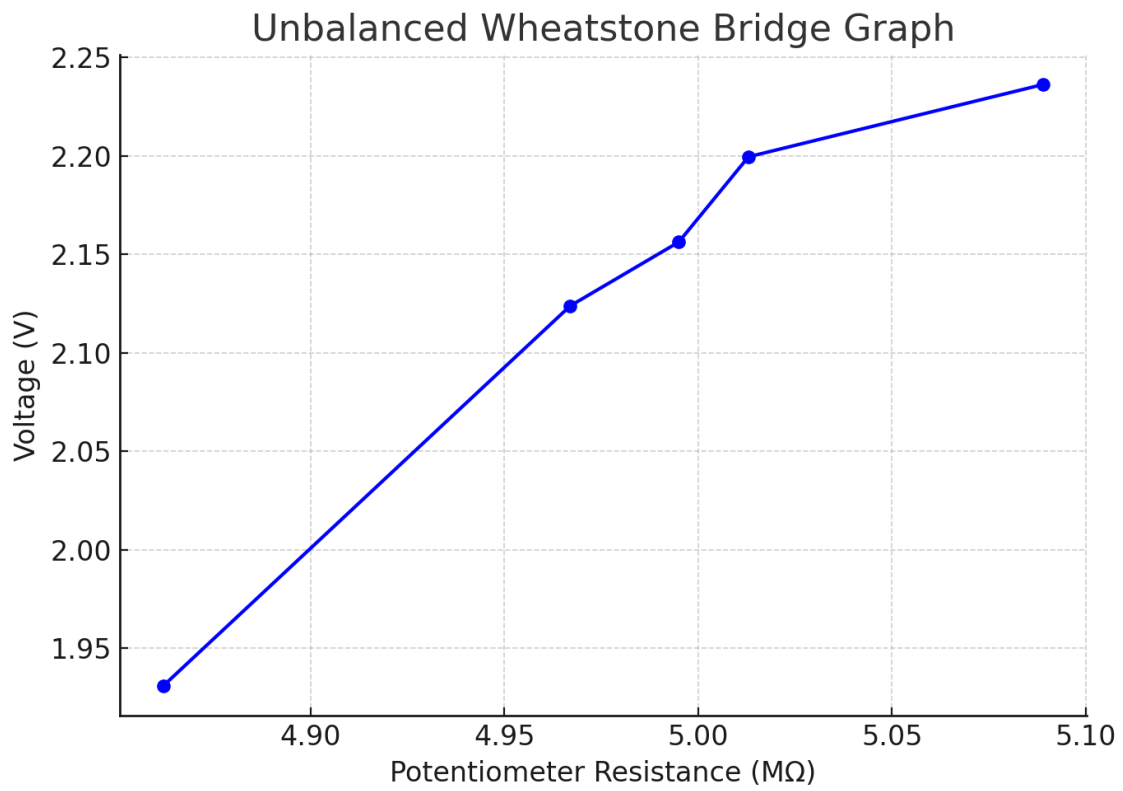


Figure 12: The obtained Graph

6 Inference

6.1 Balanced Wheatstone Bridge

- **Observation:** The galvanometer/voltmeter indicated negligible current, confirming that the bridge reached a state of balance.
- **Comparison with Theory:** The experimentally obtained resistance values closely matched the theoretical calculations, validating the accuracy of the setup.
- **Error Analysis:** Minor variations can be attributed to resistor tolerances, slight calibration issues, or measurement errors during the experiment.

6.2 Unbalanced Wheatstone Bridge

- **Observation:** A noticeable voltage appeared across the bridge points, showing that the system was in an unbalanced condition.
- **Verification:** As the potentiometer resistance was varied, the corresponding voltage readings followed the predicted trend, demonstrating good agreement between experiment and calculation.
- **Conclusion:** The unbalanced case confirmed that the bridge deviated from equilibrium, yet the experimental results were consistent with theoretical expectations.